

Network Security Analysis



ASSESSMENT TWO

Network Security Analysis

ABSTRACT

This assignment guides students through a practical and analytical exploration of network security, requiring them to capture, decrypt, and interpret HTTPS traffic using Wireshark, explain host-to-host communication at a protocol level, evaluate the security mechanisms of an HTTPS site, and perform an advanced security audit to identify potential vulnerabilities. The task blends technical execution with critical analysis and professional reporting, ensuring students develop both hands-on skills and the ability to present findings clearly and logically, preparing them for real-world network security challenges.

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Overview

This assignment applies the methods, tools, and theoretical frameworks introduced in previous tutorials to conduct an independent network security analysis. The task blends practical packet capture and analysis with structured written reporting.

You will address four interrelated questions, each focusing on a distinct facet of network security analysis. The aim is to develop your technical skills while encouraging reflective, higher-level thinking about network interactions and security mechanisms.

Submission Requirements

- Participation:** This is an individual task; you are not authorised to work in pairs or groups.
- Format:** Submit a single Microsoft Word document.
- Structure:** Clearly label and answer all four questions separately.
- File Naming:** Your file must include both your name and student ID number.
- Word Count:** Pay attention to the maximum word limits. Submissions that are unnecessarily verbose will be considered unclear and may receive lower marks.
- Generative AI:** The use of Generative AI is not authorised for inclusion in the assessment.
- Deadline:** Sunday, 5th October (23:59).

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Question 1 - Packet Capture and Data Analysis

Using what you have learned in previous lectures and Tutorials 4 and 5, you will perform your own network packet capture in this task. You will need to apply new techniques to decipher this data. Once you have done so, you will use the captured data to answer questions about the host.

Task: Using Wireshark (or an equivalent tool), perform a packet capture while using a web browser to visit: <https://www.lucanthamarineinsurance.com/>

Guidance

- a) A Linux virtual machine from your class tutorial sessions may be used, though Wireshark on Windows is often easier to configure.
- b) Because this site uses HTTPS, data will initially appear encrypted. You must configure Wireshark to decode TLS traffic to render the data intelligible. Review the concepts of HTTPS, SSL, and TLS if they are unclear.
- c) If you encounter errors loading the site, document what you did to resolve them so that traffic can be captured successfully.
- d) Use Wireshark filters to isolate relevant traffic, and include screenshots of both the encrypted and decrypted views of the capture.

Deliverable: Explain your packet capture process and data analysis methodology. (Maximum 500 words, excluding screenshots and raw packet data). Ensure that you explain the structure, the interaction sequences, and their purpose.

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Question 2 - Communication Analysis

Task: Interpret the communication taking place between your host machine and the web server.

Guidance

- a) Avoid a line-by-line packet commentary. Instead, present a higher-level explanation of the interaction between the two hosts. Discuss the role of key protocols observed in your capture, such as DNS, TLS, and HTTP(S). Ensure that you highlight any anomalies or errors in the communication flow.

Deliverable: A structured explanation of the interaction sequence and its purpose. (Maximum 500 words, excluding screenshots and raw packet data)

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Question 3 - HTTPS Security Mechanisms

Task: Produce a concise report on the security mechanisms that underpin the HTTPS connection.

Guidance

- a) Describe the technologies involved (e.g., TLS handshake, encryption algorithms, certificates).
- b) Please provide a brief historical context for HTTPS and TLS, explaining their development and significance.
- c) Evaluate the advantages and limitations of these security measures.
- d) Critically assess the configuration of this particular website, including the certificate provider's reputation and trustworthiness.

Deliverable: A formal report with analysis and observations. (Maximum 500 words, excluding screenshots and raw packet data)

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Question 4 - Advanced Security Audit

Task: Extend your audit to explore additional security aspects of the host beyond HTTPS.

Guidance

- a) Begin with an analysis of HTTP response headers and discuss what they reveal about the server configuration.
- b) Explore other services potentially running on the host, using passive reconnaissance techniques introduced in Week 3.
- c) Investigate known vulnerabilities (e.g., via CVE databases) that may affect the server or its software stack.
- d) Select two vulnerabilities (if present) to analyse in detail, discussing their severity and implications.
- e) Include screenshots of evidence wherever possible.

Deliverable: A detailed, professional discussion of findings, vulnerabilities, and insights.
(No word limit – focus on clarity and depth)

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References

All sources, including academic papers, industry standards, or any specific product documentation you consulted, must be cited correctly in a professional referencing style (e.g., Chicago 17).

Grammar and Punctuation

The report must be free of grammatical errors, spelling mistakes, and punctuation issues. A polished, error-free document demonstrates attention to detail and professional diligence.

Academic Integrity and Generative AI

This assessment requires original, independent work. The use of Generative AI tools (such as ChatGPT, Copilot, Gemini, etc.) for generating text, analysis, or content for this report is **not authorised**. All written content, data analysis, and visual materials must be the result of your own critical thinking and effort.

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Appendix

Marking Criteria (100 marks total)

Item	Marks	Related Question	Weight	Criteria
Packet Capture Technique (Q1)	15	Question 1	6	1. Screenshots of data before and after (6 marks)
			4	2. Wireshark filters used (4 marks)
			5	3. Explanation of technique and assumptions (5 marks)
Explanation of Packets (Q2)	20	Question 2	10	1. Commentary on what is taking place (10 marks)
			4	2. Appropriate filtering and screenshots (4 marks)
			6	3. Comprehensive identification and discussion (6 marks)
Report on HTTPS (Q3)	15	Question 3	5	1. General overview of the technology (5 marks)
			6	2. Logical flow and clear discussion (6 marks)
			4	3. Specifics of the target site (4 marks)
Advanced Audit (Q4)	30	Question 4	10	1. Discussion of extracted HTTP data (10 marks)
			5	2. Includes new sources of information (5 marks)
			5	3. Highlights noteworthy findings (5 marks)
			5	4. Explains the value of the information (5 marks)
			5	5. Discusses known vulnerabilities (5 marks)
Presentation	20	Document format, style, etc.	5	1. Clear, logical flow (5 marks)
			5	2. No spelling or grammatical mistakes (5 marks)
			5	3. Sections are numbered and labelled (5 marks)
			5	4. Report is finished to a professional standard (5 marks)

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Suggested Student Report Template

Although the presentation of the document is entirely the choice of the student, we would suggest that you consider a more professional presentation, similar to the structure below, for your submission.

1. Header Page
 - Assignment title
 - Unit code
 - Student name and ID
 - Date of submission
2. Table of Contents
 - Include page numbers and clear headings.
3. Section 1 – Packet Capture & Deciphering (Q1)
 - Introduction
 - Environment & Tools Used
 - Step-by-Step Methodology
 - Screenshots that should include a before and after decryption. You could also add text to the screenshots for clarity.
 - Key Observations
4. Section 2 – Host Communication Analysis (Q2)
 - Overview of Communication Flow
 - Protocol-Level Discussion (DNS, TLS, HTTP/S)
 - Summary of Findings
5. Section 3 – HTTPS Security Mechanisms (Q3)
 - Overview of HTTPS and TLS
 - Historical and Technical Context
 - Advantages & Limitations
 - Site-Specific Observations
6. Section 4 – Advanced Security Audit (Q4)
 - HTTP Header Analysis

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- Other Service & Capability Findings
 - Vulnerability Research and Discussion
 - Evidence Screenshots
7. Conclusion
 8. Summary of insights
 9. Reflection on security posture
 10. References
 - Follow Chicago's style for citations.